

the same return. Additionally, the following table compares the relevant moments of the empirical and simulated wealth distributions.

**Table 1.2:** Wealth Distribution (PY)

|                               | <b>0–20</b> | <b>20–40</b> | <b>40–60</b> | <b>60–80</b> | <b>80–95</b> | <b>Top 5%</b> |
|-------------------------------|-------------|--------------|--------------|--------------|--------------|---------------|
| <b>Wealth share (data)</b>    | -0.002      | 0.011        | 0.044        | 0.118        | 0.255        | 0.574         |
| <b>Wealth share (no het.)</b> | 0.030       | 0.085        | 0.142        | 0.224        | 0.285        | 0.233         |
| <b>Wealth share (het.)</b>    | 0.005       | 0.019        | 0.042        | 0.096        | 0.251        | 0.588         |

#### 1.4.2 Incorporating life cycle dynamics into the model

Assumptions about the age and education level of households can have important implications for the income and mortality process of households. Next I extend the model to incorporate these life-cycle dynamics.

Households enter the economy at time  $t$  aged 24 years old with an education level  $e \in \{D, HS, C\}$ , an initial permanent income level  $\mathbf{p}_0$ , and a capital stock  $k_0$ . The life cycle version of household income is given by:

$$y_t = \xi_t \mathbf{p}_t = (1 - \tau) \theta_t \mathbf{p}_t,$$

where  $\mathbf{p}_t = \psi_t \bar{\psi}_{es} \mathbf{p}_{t-1}$  and  $\bar{\psi}_{es}$  captures the age-education-specific average growth factor. Households that have lived for  $s$  periods have permanent shocks drawn from a lognormal distribution with a mean of 1 and a variance of  $\sigma_{\psi_s}^2$  and transitory shocks drawn from a lognormal distribution with a mean of  $\frac{1}{\bar{\psi}}$  and a variance of  $\sigma_{\theta_s}^2$  with probability  $\bar{\psi} = (1 - \bar{\psi})$  and  $\mu$  with probability  $\bar{\psi}$ .